Question Bank

Unit 2

Q.1 Construct Regular Expression for given ~~Grammar~~ Languages.

i) Language with No string { }
ii) Language with length 0 {ε}
iii) Language with length 1 {a, b}
iv) Language with length 2 {}
v) Language with length 3
vi) Language with atleast 2
vii) Language with atmost 2.
viii) All strings having ~~b~~ single 'b'
ix) All strings having 'bbbb' as substring
x) All strings containing two b's
xi) All string start with 'ba'
xii) All string end with 'ab'
~~xiii~~ 13) All string start with 'a' and end with 'a'
14) Start with a and end with b
15) string catching 'a'

Q.2. What are the properties of Regular Expression in TOC

Q.3. What are the Restrictions of Regular Grammar

Q.4. Prove the equalities of Regular Expression by applying properties

Q.5 State any four closure properties of Regular set.

Q.6. Using Pumping Lemma Prove that the language $A = \{0^n 1^n \mid n \geq 0\}$ is not Regular

Q.7 Using pumping Lemma Prove that the language $A = \{a^n b^n \mid n \geq 0\}$ is not Regular.

Q.8 Elaborate the term Regular Grammar. with types & examples.

Q.9. Elaborate LLG, RLG with example.

Q.10 Elaborate any Four properties of Regular set.

Q.11 convert the following LLG to RLG

$S \to Aa \mid Bb$

$A \to Bb$

$B \to Ba \mid b$

Q.12. Generate the Language from given Grammar

$G = (\{S, A, B\}, \{a, b\}, S, \{S \to AB, A \to aA \mid a, B \to bB \mid b\})$

Q.13 Generate the Language from given Grammar

1) $G1 = (\{S, A\}, \{a, b\}, S, \{S \to aAb, aA \to aaAb, A \to \epsilon\})$.

2) $G2 = (\{S, A, B\}, \{a, b\}, S, \{S \to AB, A \to a, B \to b\})$.

Q.14 Convert the following RLG to LLG.

$S \to aA$

$A \to bA \mid b \mid a$

Q.15 convert the following LLG to RLG

1) $S \to Ca \mid Aa \mid Bb$

$A \to Ab \mid Ca \mid Bb \mid a$

$C \to Aa$

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